

Cloud Computing with AWS

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Plan

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- Networking
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Introduction

Drawbacks of legacy IT infrastructure

- Figure out in advance the requirement of each application
- High Upfront cost to buy and install the infrastructure
- Cooling and electricity expenses
- Maintenance and update fees
- Limited capacity and/or underutilized servers
- Stuck with these resources for many years

What is cloud computing?

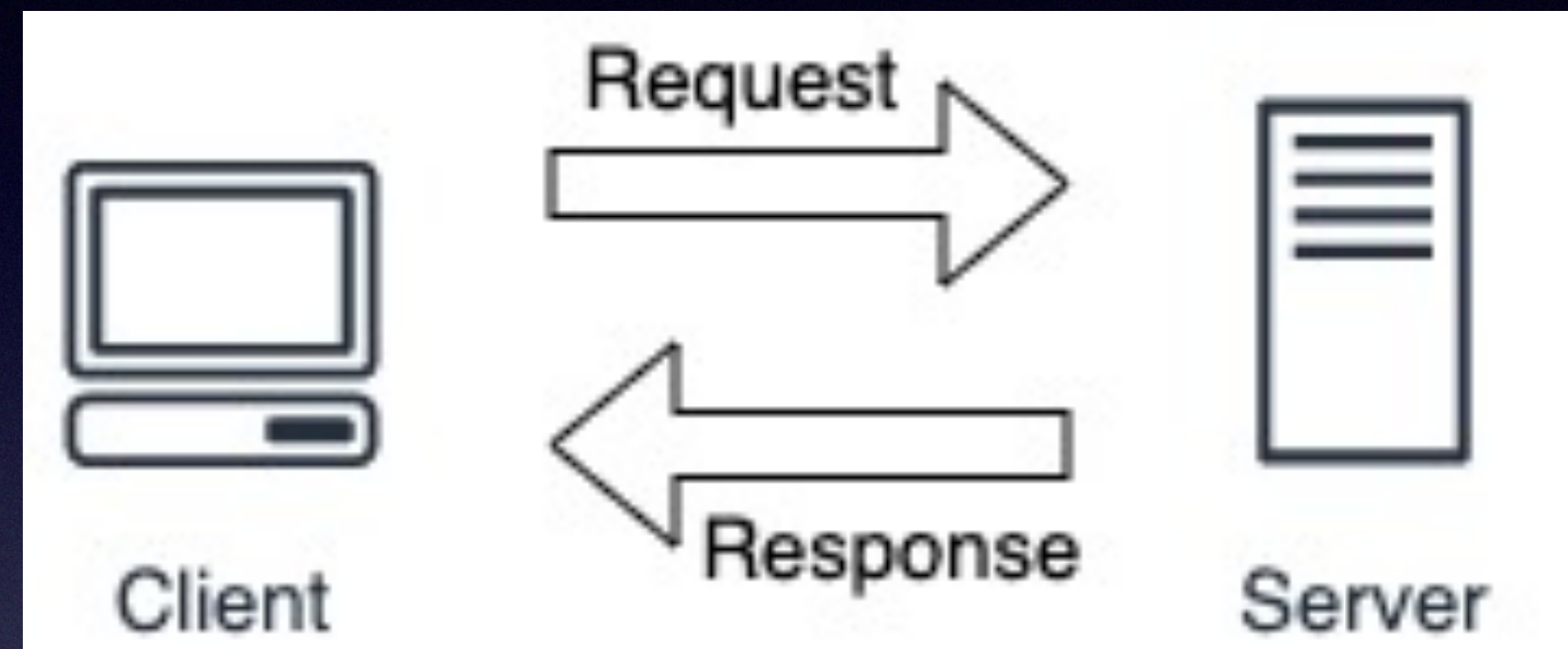
On demand delivery of IT resources over the internet with pay as you go pricing.

- On demand delivery: get resources when the user asks for them.
- IT resources: compute and storage resources and services.
- Over the internet: the resources are remotely accessible from a browser or programmatically.
- Pay as you go pricing: the user only pays when using resources.

What can be hosted in the cloud?

- Websites (static or dynamic)
- Server side of mobile Apps (backend)
- IT services (warehouse or payroll management systems)
- Video streaming (Netflix, Amazon prime)
- Online video games (luna, game pass, geforce now, etc.)
- Scientific simulation and big data analysis

Client/Server model



- The user of the cloud is the client and the cloud is the server.
- The user requests services from the cloud (execute tasks, store data, etc.) and pays for these services.
- The cloud immediately fulfills the requests of the user.

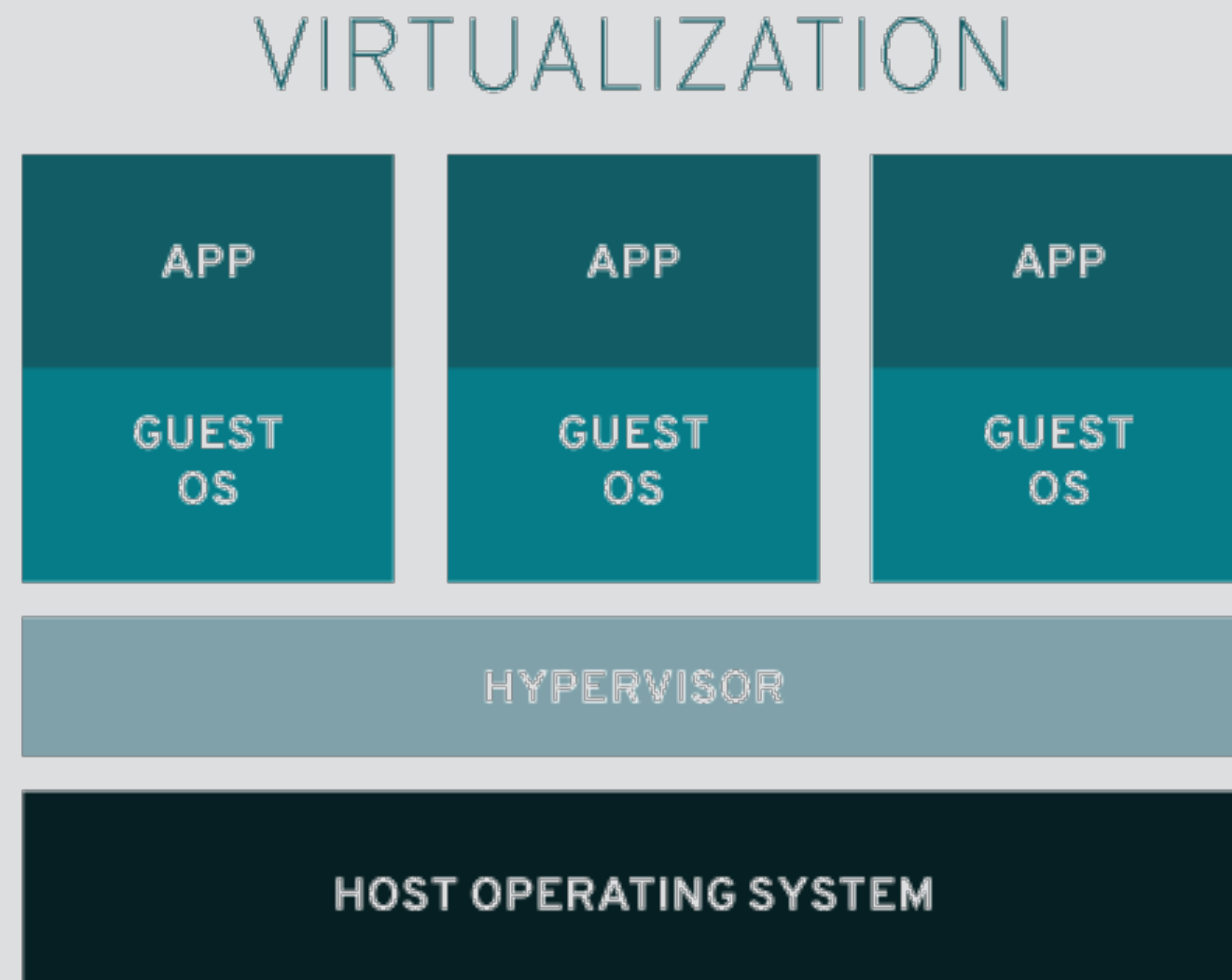
Characteristics of cloud computing

- Virtualization technology allows multiple users to share a host with complete isolation between them.
- Managed services are operated and maintained by the cloud provider.
- Services are available on-Demand. Payment for only the used resources.
- Networks can be customized and isolated for security purposes.
- Scalability & Elasticity refer to the ability of cloud resources to grow or shrink on-demand.

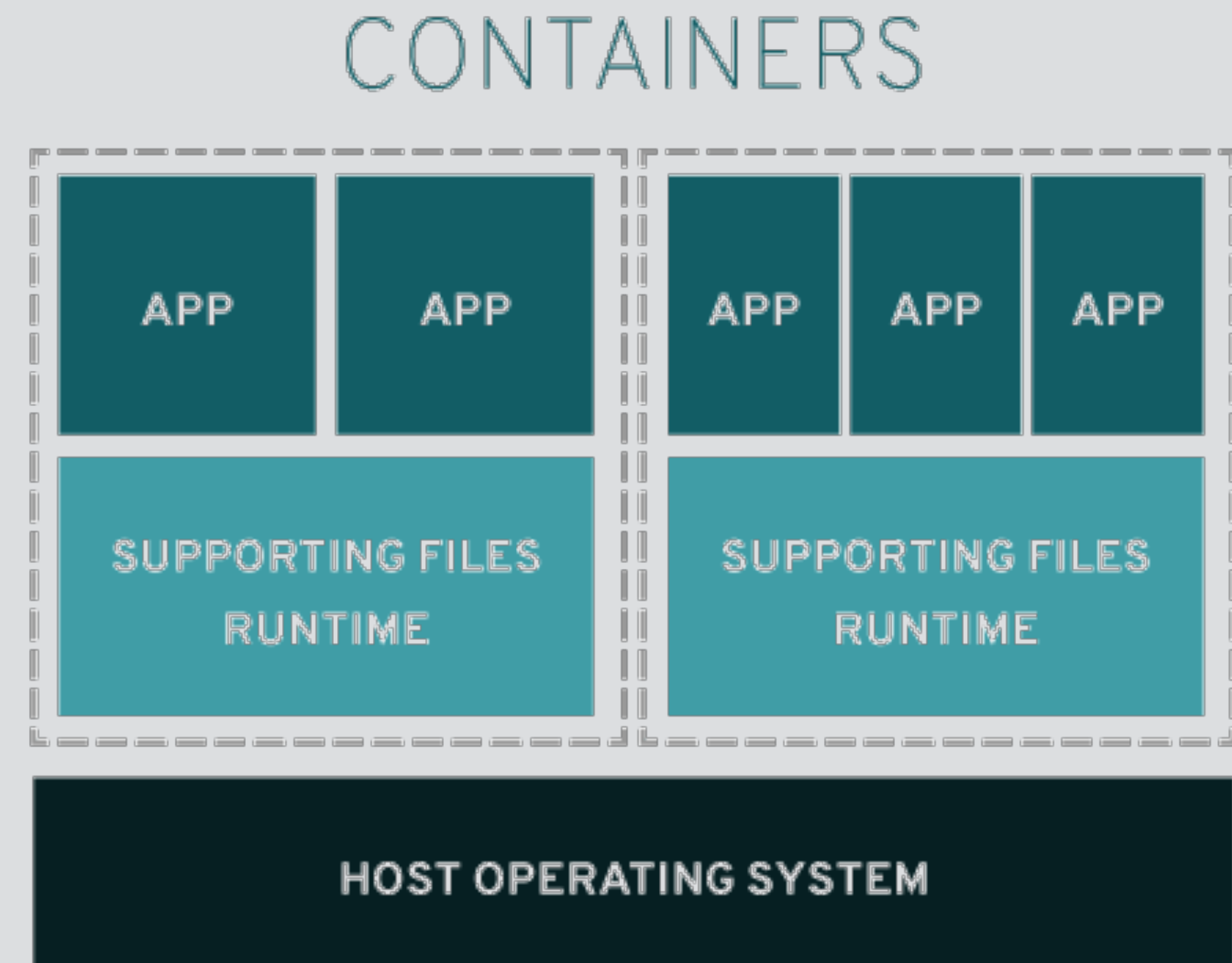
Virtualization

- Allows multiple VMs to run on a single physical server, Multi-tenancy.
- The physical server, the host, runs a hypervisor to manage the VMs.
- A Virtual Machine (VM), guest, has its own OS.
- The host allocates a configurable amount of its resources to each VM
- The hypervisor isolates the VMs from each others. Each VM is not aware of the other VMs running on the same host.

Virtualization vs containers



VS.



Deployment models

1. Private cloud: similar to legacy IT infrastructure where the resources and services are deployed on premises. Management tools and virtualization technologies are used to manage those resources and increase their utilization.
2. Public cloud: the cloud provider such as AWS, Azure or GCP, provides the resources and services. The user just pays for the resources he is using. No upfront expenses and maintenance fees.
3. Hybrid cloud: a mix of the two previous models. It benefits from the privacy and security of a private cloud to process confidential data while using public cloud for more compute or memory intensive applications.

Benefits of public clouds

1. No upfront expenses (servers, network, cooling system, etc.)
2. No maintenance fees (hardware and software)
3. Stop guessing capacity
4. Massive economies of scale
5. Increase speed and agility (access new resources within minutes)
6. Deliver content to customers around the world with minimal delay

Types of cloud computing

- Infrastructure as a service (IaaS) allows users to rent resources on-demand instead of having to buy and maintain the hardware.
- Platform as a service (PaaS) allows developers to build upon a framework to create custom applications. By taking care of provisioning, deployment etc. PaaS providers help users focus on their applications, rather managing the base infrastructure.
- Software as a service (SaaS) allows companies to deliver applications that run in the browser, without any installation.

Major cloud providers

- AWS 32%, Azure 22% and GCP 11%.
- Factors to consider when choosing a cloud provider:
 - Geographic Availability (Azure has the most regions, 60+)
 - Performance
 - Services (AWS has the most services, 200+)
 - Pricing Model (Hourly pricing, license pricing, CPU usage, etc.)

Interacting with AWS

How to interact with AWS?

- There are predefined APIs to interact with the AWS services that allow the user to provision, configure, and manage AWS resources.
- The user can use the AWS Management Console, the AWS Command Line Interface, the AWS Software Development Kits, or various other tools like AWS CloudFormation to create requests to send to AWS APIs to create and manage AWS resources.

AWS management console

- It's a Web interface to visually manage the AWS resources. It is the easiest way for getting started.
- The console includes wizards and automated workflows that can simplify the process of completing tasks.
- It's also useful for viewing AWS bills or working with non-technical resources. There is also a mobile app.
- Point and click style is not fit for production type environment. Manual configuration for each service may lead to errors.

AWS CLI and SDKs

- To automate the interaction with AWS services, scripting or programming the API calls through the Command Line Interface (CLI) or a Software Development Kit (SDK) should be used.
- Reduce the errors due to manual interaction.
- Increase the productivity of the user.
- CLI is available for users on Windows, macOS, and Linux.
- There are SDKs for various languages such as C++ and Java.

CLI commands to manage S3

- Create a bucket: `aws s3 mb <target> [--options]`
- List buckets and objects: `aws s3 ls <target> [--options]`
- Remove bucket: `aws s3 rb <target> [--options]`
- Copy objects: `aws s3 cp <source> <target> [--options]`
- Sync objects: `aws s3 sync <source> <target> [--options]`

AWS Elastic Beanstalk

- Instead of clicking around the console or writing multiple commands to build out the services, the user can provide the application code and desired configurations to AWS Elastic Beanstalk, which then builds out the environment to run the code.
- AWS Elastic Beanstalk also makes it easy to save environment configurations, so they can be deployed again easily. The user gets to focus on the business application, not the infrastructure.

AWS CloudFormation

- It helps the user to achieve automated and repeatable deployments.
- It is an infrastructure as code tool, that allows the user to define a wide variety of AWS resources in a declarative way using JSON or YAML text based documents, called CloudFormation Templates.
- CloudFormation lets the user define what he wants, and the CloudFormation engine will call the API's to get everything built out.
- The user can run the same cloud formation template on multiple accounts or regions, and it will create identical environments across them. There is less room for human error as it is a totally automated process.

CloudFormation JSON template

Template Anatomy

```
{
  "Description" : "Create an EC2 instance.",
  "Resources" : {
    "Ec2Instance" : {
      "Type" : "AWS::EC2::Instance",
      "Properties" : {
        "KeyName" : "my-key-pair",
        "ImageId" : "ami-75g0061f",
        "InstanceType" : "m1.medium"
      }
    }
  }
}
```


Access Management

Identity and Access Management

- It manages access to AWS resources through two main operations: authentication and authorization.
- In an AWS account, users and roles can be created through the IAM service.
- Policies can also be defined to specify which resources a user can access through a set of permissions.
- Users should be granted the least permissions they need.

Users and policies

- An IAM user is an identity that allows access to AWS resources, either via the console (username and password authentication) or programmatically (access key and secret key authentication).
- IAM permissions are granular and granted via "policies". By default all the permissions are denied.
- There are three types of IAM policies:
 - "Built-in" policies provided by AWS
 - "Custom" policies created by AWS users
 - "Inline" policies embedded in a IAM entity

Policy

IAM policies are formatted as JSON documents:

- Effect: allow or deny
- action: permissions, format: resource:action
- ARN: Amazon resource name uniquely identifies a resource and its sub-resources.



```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "s3:GetObject",
9         "s3:GetObjectVersion"
10      ],
11      "Resource": "arn:aws:s3:::demo-s3-talend/*"
12    }
13  ]
14 }
```


Role

- Role is similar to a user however instead of being uniquely associated with one person, a role is intended to be assumable by anyone who needs it.
- It does not have standard long-term credentials such as a password or access keys associated with it. Instead, when a user assumes a role, it provides him with temporary security credentials.
- Policies can be added to roles to give them access to resources.
- A role is usually given to trusted services to allow service-to-service communication.

Compute services

Elastic Compute Cloud (EC2)

- It's the compute service of AWS that allows the user to reserve virtual servers on which the user will run its applications through VMs.
- Get access in minutes to a large number of virtual servers with no upfront or maintenance expenses.
- Pay only for the time the virtual servers are used and release them when done.
- The user can vary his usage according to his needs.
- Virtual servers run on top of physical hosts using virtualization technology.

EC2 Instance

- An EC2 instance is a VM running on a virtual server in the cloud.
- The user can configure a VM before booting it:
 - Choose the operating system: Windows or Linux.
 - Choose the applications running on the VM: web server, DB, etc.
 - Choose the resources of the VM in terms of compute power, memory, disk size, etc. They can be modified through vertical scaling.
 - Configure the network of the VM: is it publicly available?

EC2 Instance types

- Each Instance type has a predefined combination of CPU, memory, storage and network capacities.
- Each Instance type belongs to an Instance family:
 - general purpose: balanced resources
 - compute optimized: high performance procs for compute intensive apps
 - memory optimized: large memory to process big data
 - storage optimized: fast local storage for I/O intensive apps
 - accelerated computing: include accelerators such as GPUs and TPUs

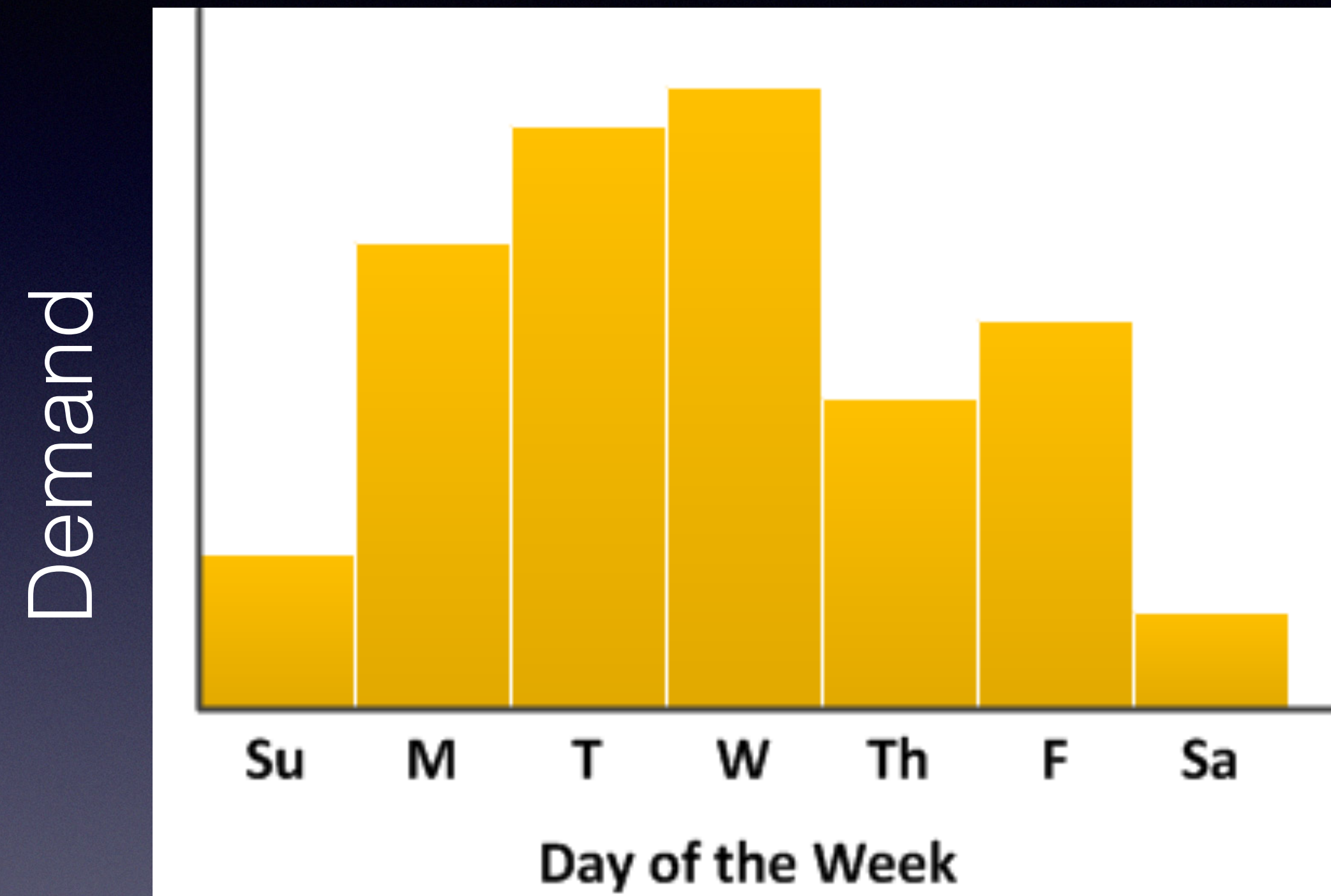
EC2 pricing models (1/2)

- On-demand: only pay for the duration your instance runs for.
Advantages: no longtime commitment or upfront costs. It is ideal for deploying new apps in the cloud and evaluating its resource usage.
- Savings plans: Savings plan offers low prices on EC2 usage in exchange for a commitment to a consistent amount of usage.
Measured in dollars per hour for a one or three year term.
- Reserved instances are suited for steady state workloads where the client commits for one or three years to using a given instance in order to get discounts.

EC2 pricing models (2/2)

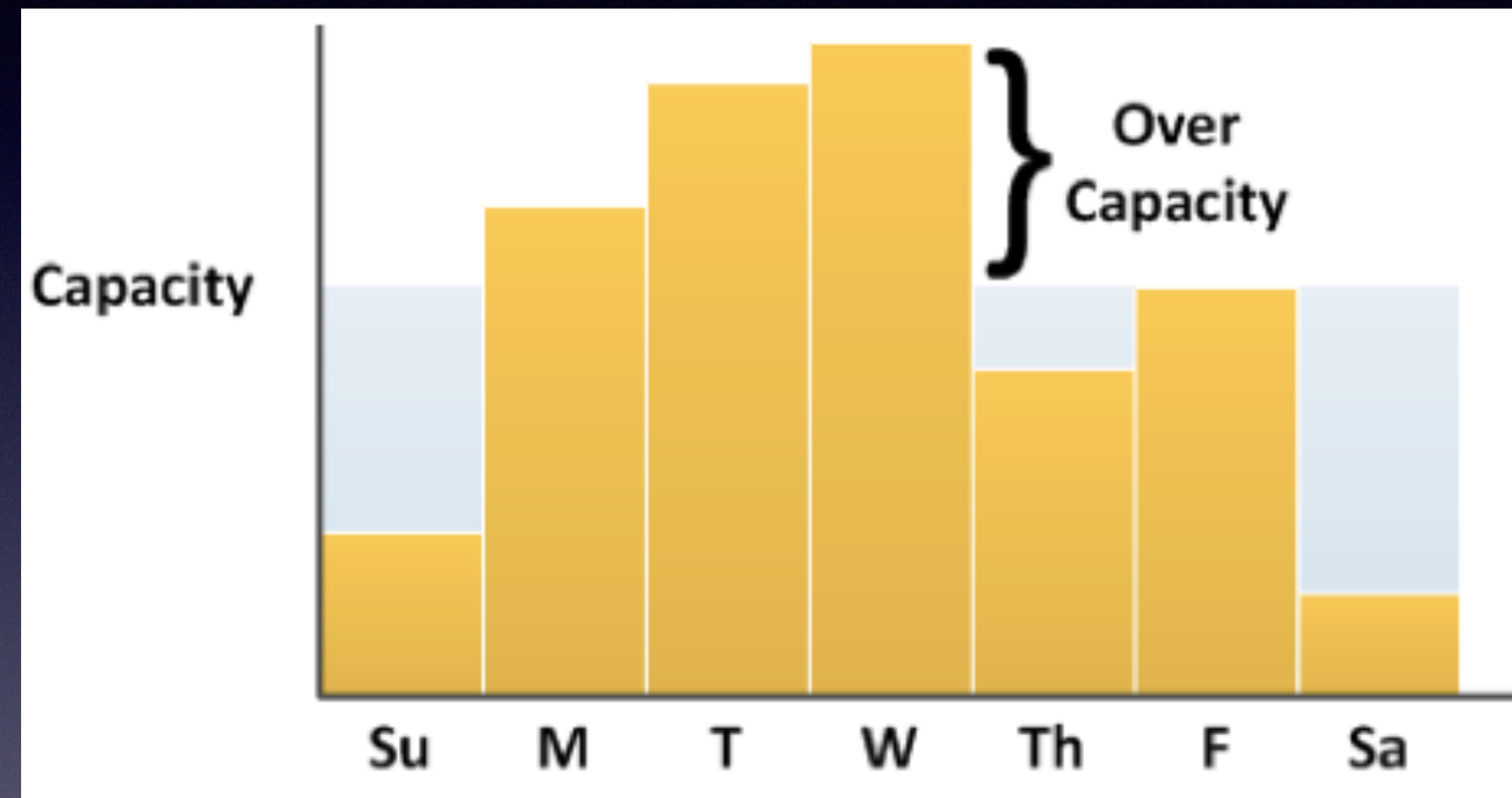
- Spot instances allows the user to save up to 90% of the on demand price. However AWS can reclaim the instance at any time they need it. Giving the user a two minutes warning to finish up work, and save the state of the application.
- Dedicated hosts are physical hosts dedicated for the user. The physical hosts are not shared with other users to meet certain compliance requirements such as confidentiality, security, etc. These are the most expensive.

EC2 scaling



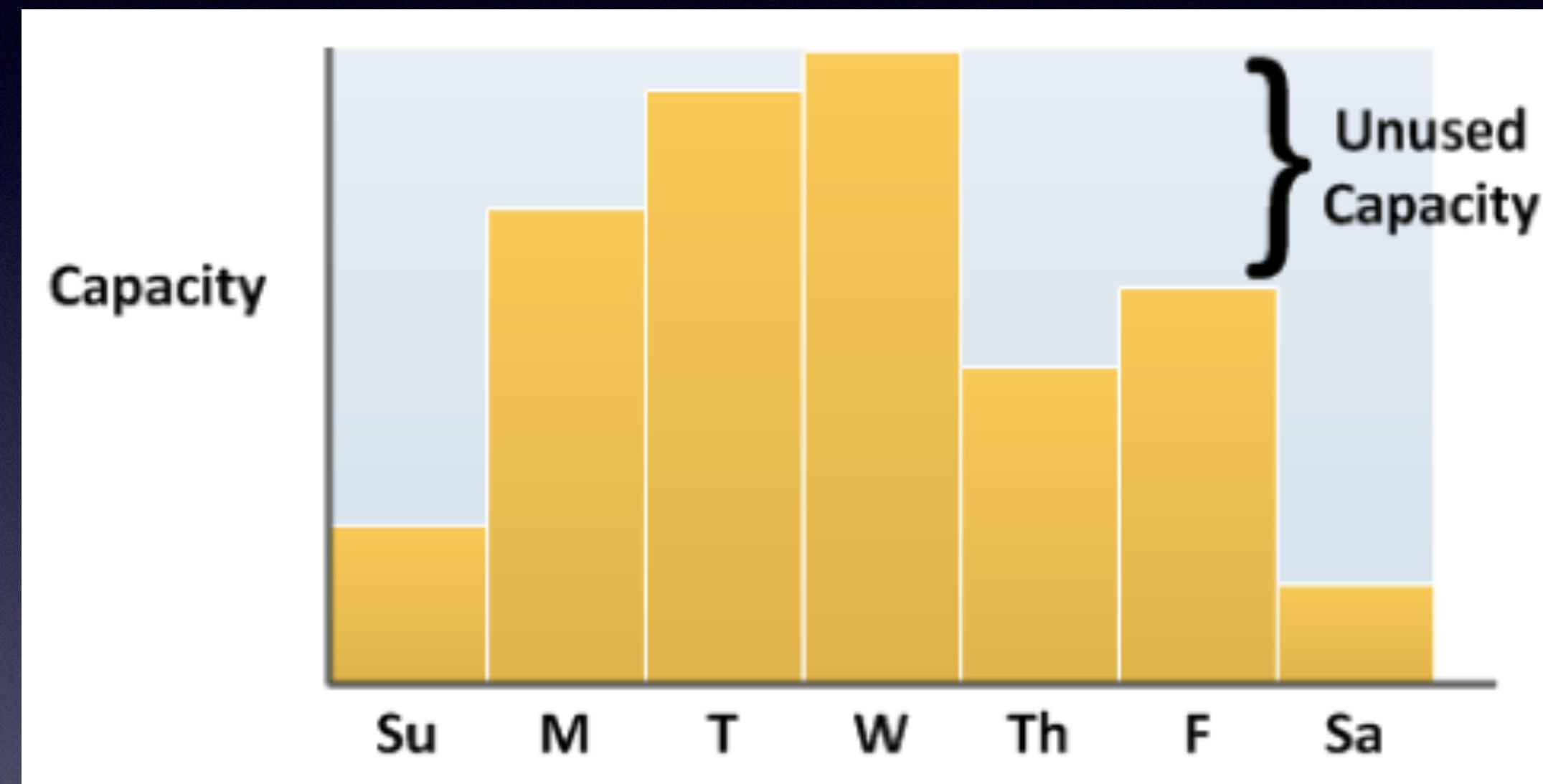
Which capacity is the most suited for such variable demand?

EC2 scaling



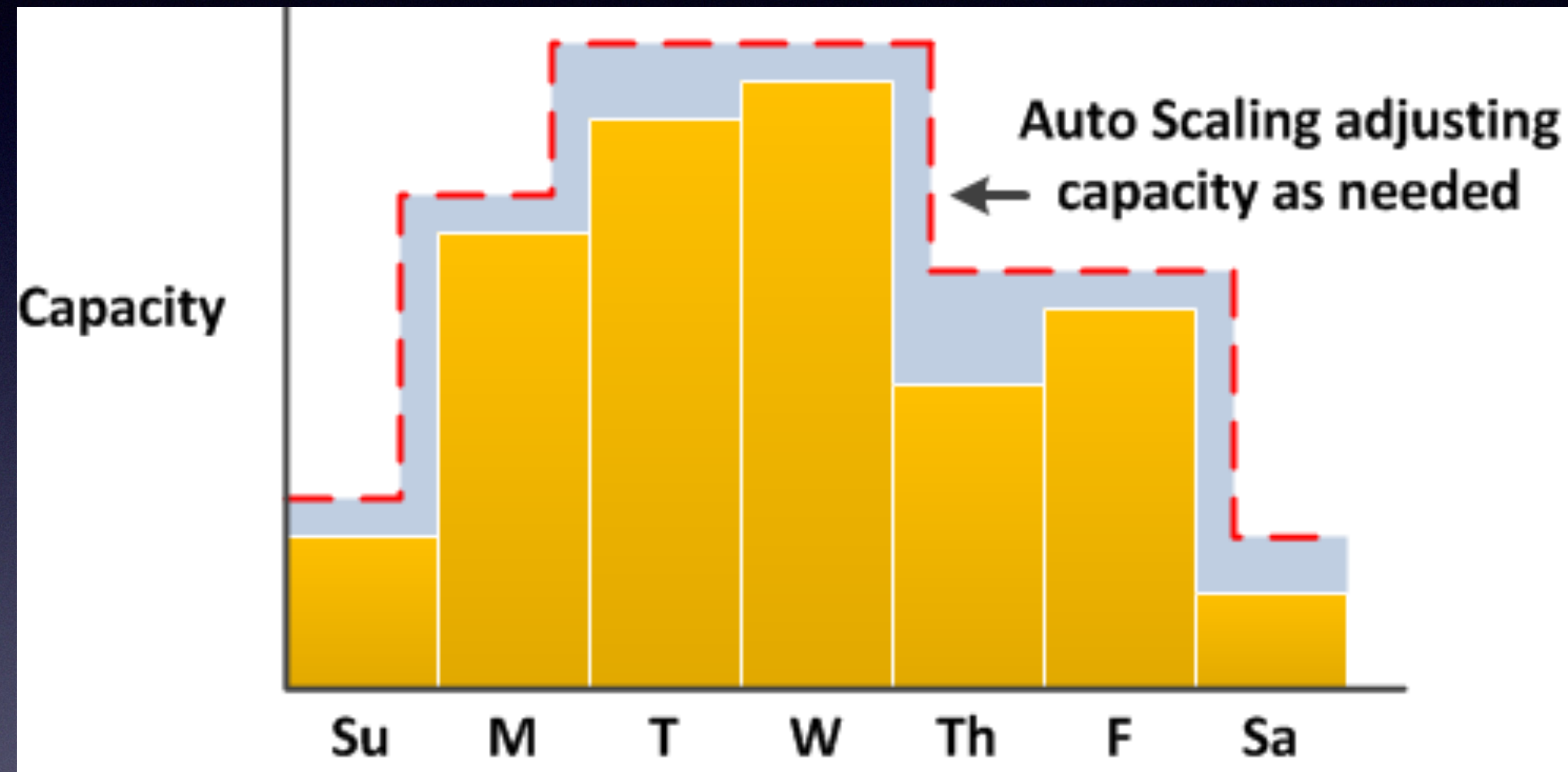
No sufficient resources to serve all the customers during peak hours, days or seasons.

EC2 scaling



Can handle peak demands but the resources are underutilized most of the time.

EC2 scaling



Provision enough resources to exactly meet the demand every day.

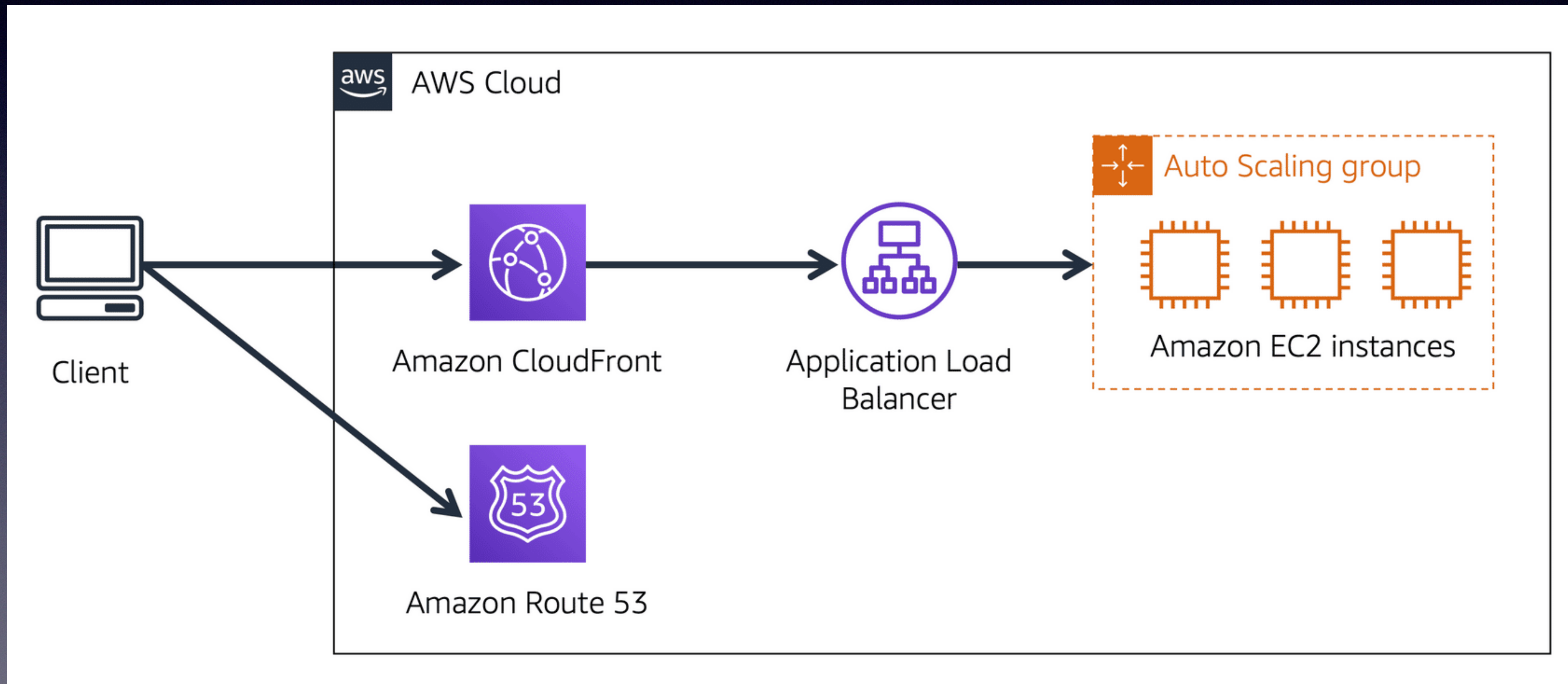
EC2 Auto scaling

- Auto scaling involves automatically responding to changing demand by scaling out or in (adding or removing EC2 instances). The user only pays for the resources he used and he doesn't have to worry about a lack of computing capacity to meet the customers' needs.
- With EC2 Auto Scaling, two approaches can be used:
 - Dynamic scaling responds to changing demand.
 - Predictive scaling automatically schedules the right number of EC2 instances based on predicted demand.

Load balancing

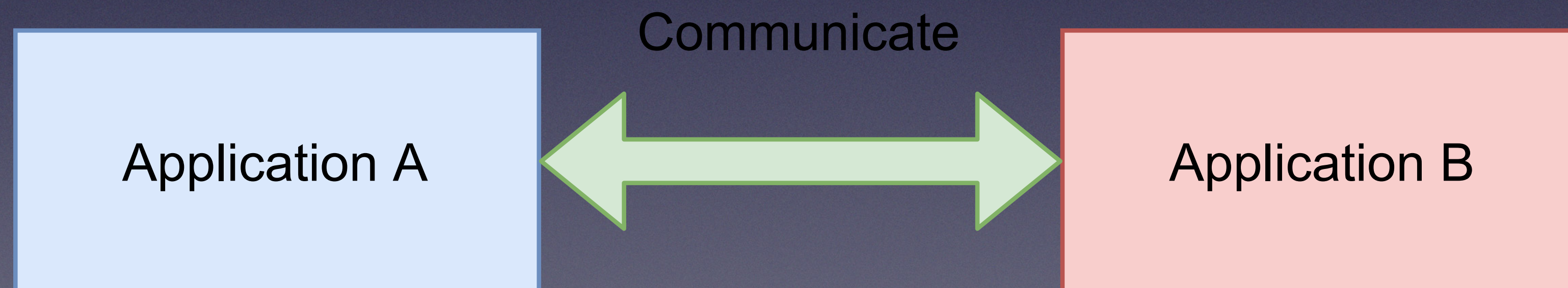
- To use all the provisioned instances, a load balancer should route the incoming workloads to the least loaded instance.
- The workloads are not directly sent to the instances. They are sent to a load balancer which transfers them to an active instance. Only the load balancer needs to know if the number of instances was scaled down or up.
- AWS proposes Elastic Load Balancing (ELB) which is highly available and scalable with no additional effort or hourly cost.

Accessing a website hosted on AWS



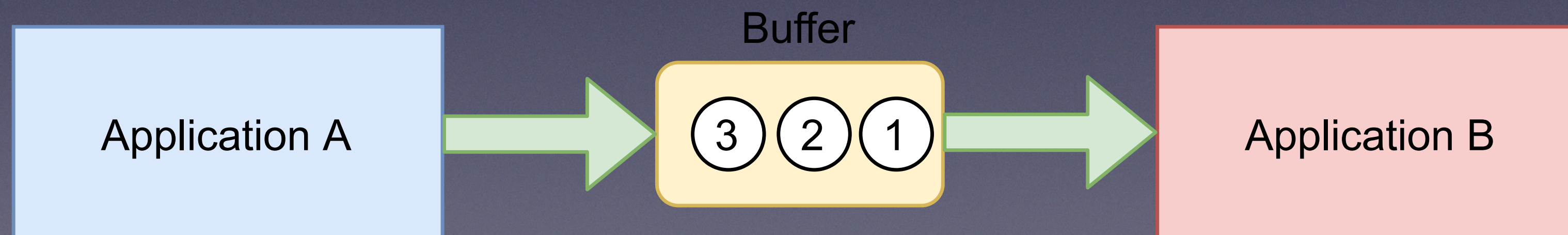
Multi-component applications

- They are composed of separated applications that communicate between each other to exchange data.
- Advantages: Reusability, Separating functionalities, easier to maintain, etc.



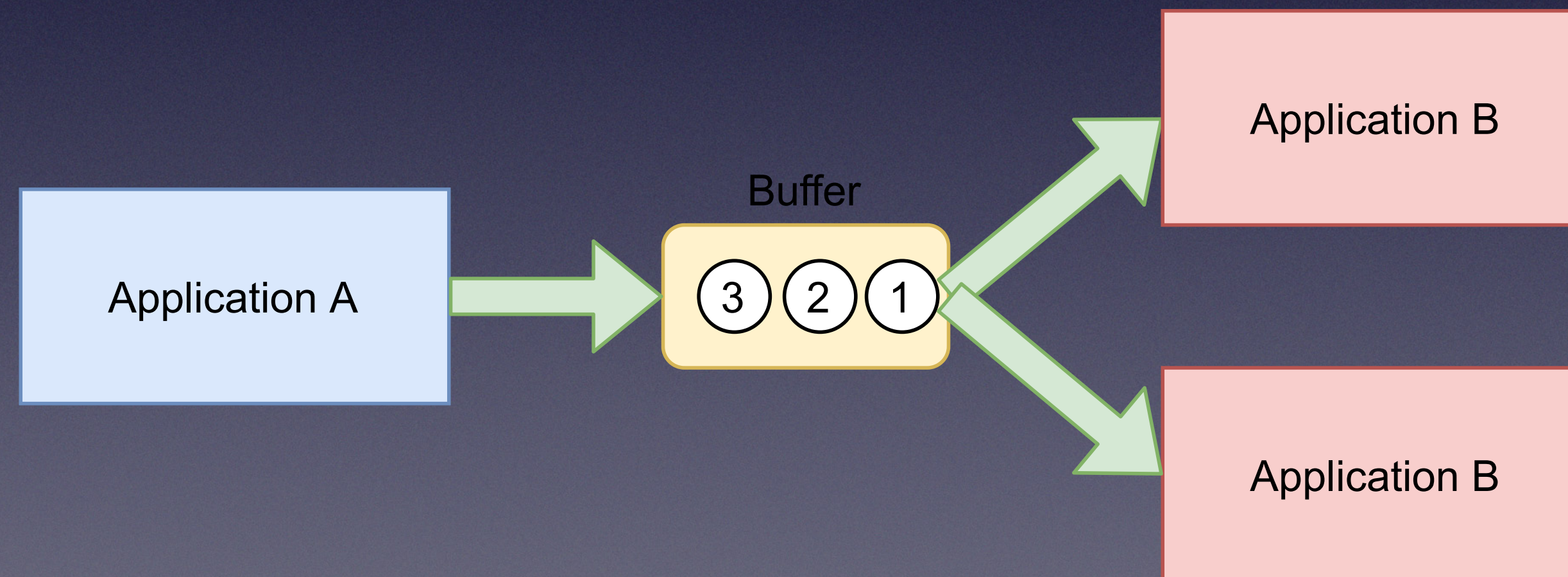
Simple Queue Service (SQS)

- Tightly coupled applications VS loosely coupled applications
- SQS is a scalable queuing service
- No need for synchronization, no cascading failures, scalable apps



Simple Queue Service (SQS)

- No single point of failure thanks to redundancy
- High availability



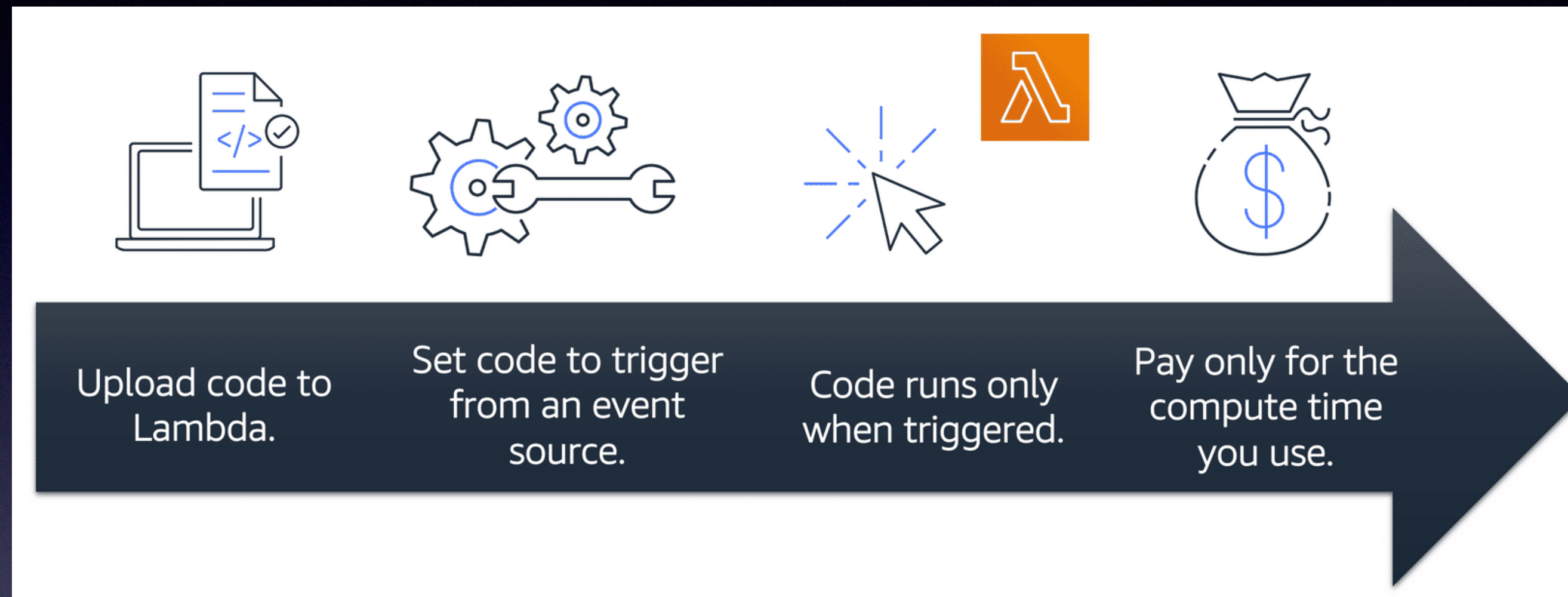
Drawbacks of EC2

- It requires that the user sets up and manages his fleet of Instances over time:
 - The user is responsible for patching his Instances when new software packages come out.
 - It is up to the user to set up the scaling of his Instances, as well as ensuring that they are highly available.
- Alternatives to EC2: Serverless compute options.

Serverless compute

- The user cannot access the underlying infrastructure or Instances, that are hosting his application.
- AWS take care of all the management of the underlying environment from a provisioning, scaling, high availability, and maintenance perspective. The user needs to focus only on his application.

AWS Lambda



- The user doesn't have to manage the environment because it is automatically scalable, highly available, and all of the maintenance is done by AWS.
- Lambda is designed to run code under 15 minutes.

Docker container-based workloads

- Docker uses operating system level virtualization to deliver software in containers where the application, its dependencies, as well as its configurations are packaged.
- The containers run above EC2 instances in isolation from each other and can be managed with orchestration tools such as Elastic Container Service (ECS) or Elastic Kubernetes Service (EKS).
- If the user doesn't want to manage his EC2 Instances to run containerized apps, he can use AWS Fargate which is a serverless compute platform for ECS or EKS.

AWS global infrastructure

AWS global infrastructure

- To ensure high availability and fault tolerance, AWS deployed data centers all over the globe especially closest to the business traffic demands.
- It divided the globe into regions and each region contains at least two data centers, ex: Paris, Oregon, Cape town, etc.
- Regions are interconnected through a high speed fiber network controlled by AWS, but they are isolated from each other.
- The User chooses in which region to deploy his application.

Selecting a region

Based on which factors should a user select a region:

- ▶ Compliance with national regulations
- ▶ Proximity to customers
- ▶ Availability of features/services
- ▶ Pricing

Availability Zone (AZ)

- Each AZ is one or more data centers.
- Each AWS region consists of multiple, isolated, and physically separate availability zones within a geographic region.
- AZs are not built next to each others but close enough to have low latency
- To avoid disaster, a cloud application should be run on at least two EC2 instances in two distinct AZs.
- Many of the AWS services run at the region level such as ELB.

AWS Edge Locations

- If some customers are in Paris and the data is hosted out of the Tokyo region, rather than having all the Paris-based customers send requests all the way to Tokyo to access the data, a copy can be placed in Paris.
- Edge location is a site that Amazon CloudFront uses to store cached copies of the user's content closer to his customers for faster delivery.
- Edge locations are separate from regions, the content can be pushed from inside a region to a collection of Edge locations around the world in order to accelerate communication and content delivery.

Networking

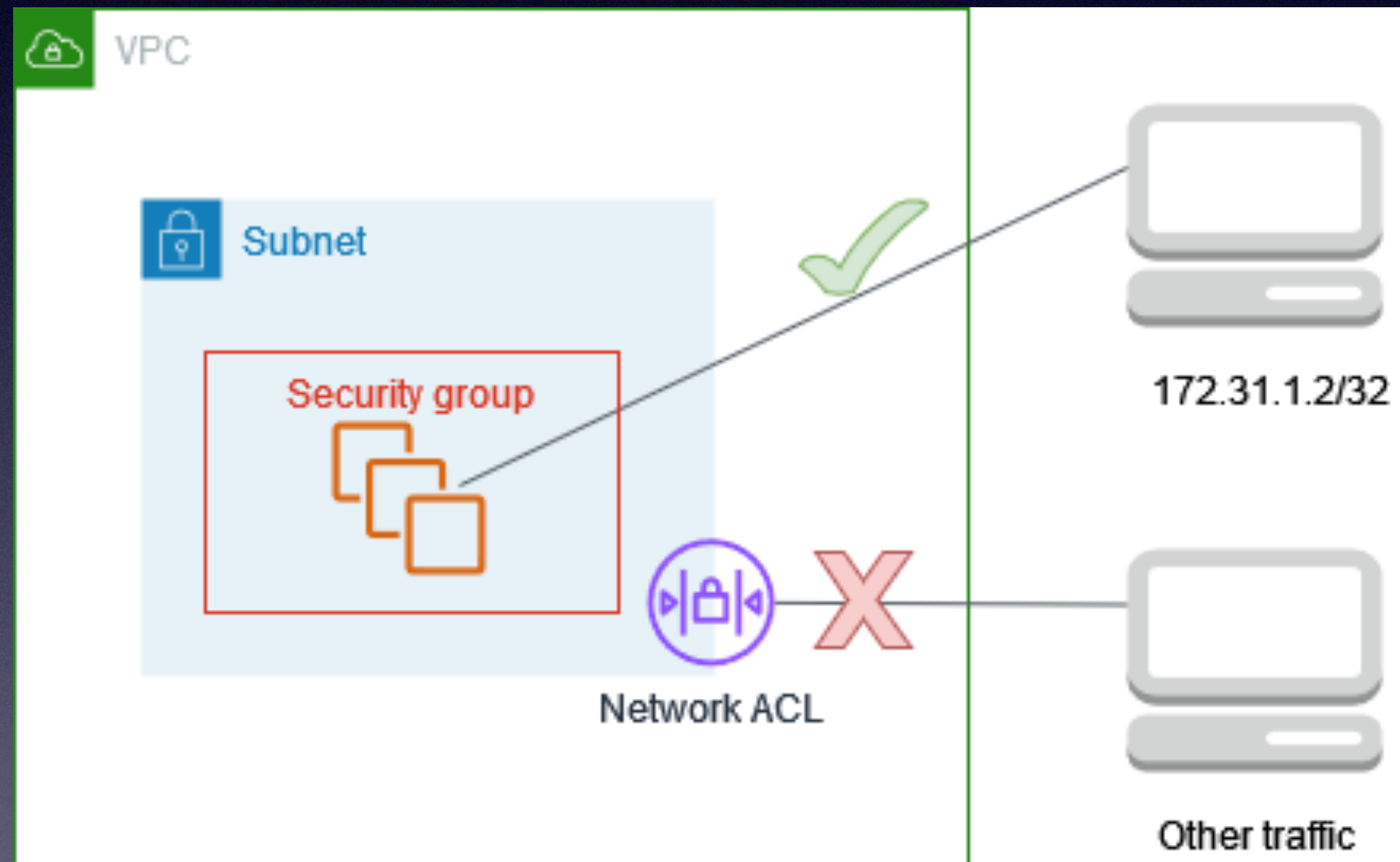
Virtual Private Cloud (VPC)

- A VPC lets the user provision a logically isolated section of the AWS cloud where he can launch AWS resources in a virtual network.
- These resources can be public-facing so they have access to the Internet (websites) or private with no Internet access (back-end services like databases or application servers).
- A VPC can be divided into subnets which are chunks of IP addresses in your VPC.
- Subnets along with networking rules control whether resources are either publicly, or privately available.

Network Access Control List

- Network Access Control List (ACL) checks traffic going into and leaving a subnet. Approved traffic can be sent on its way and the rest gets blocked.
- Network ACL only checks packets at a subnet's boundary. It doesn't evaluate if a packet can reach a specific EC2 Instance or not.
- By default, Network ACL allows all inbound and outbound traffic.

Example of ACL rules

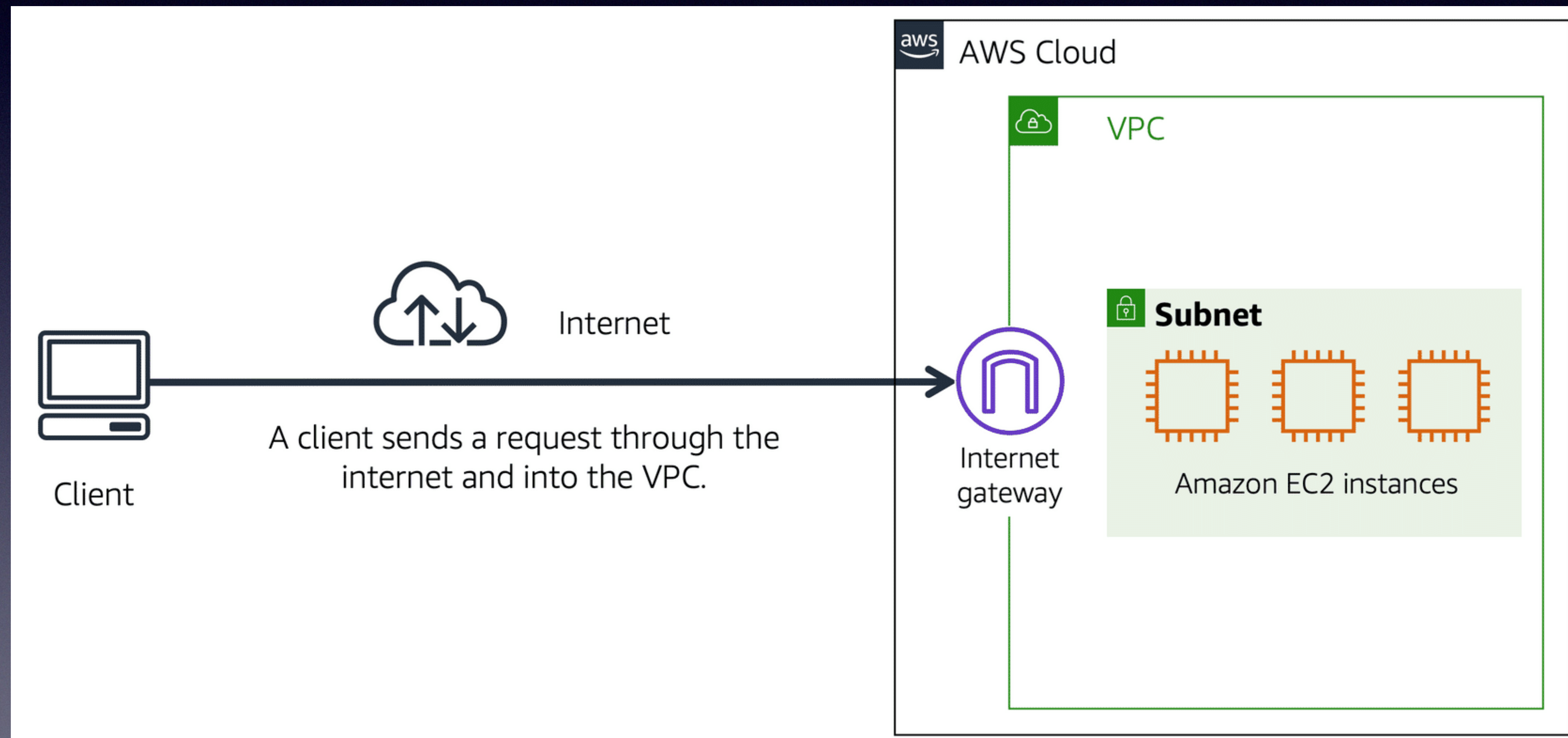


Rule #	Type	Protocol	Port range	Source	Allow/Deny	Comments
100	SSH	TCP	22	172.31.1.2/32	ALLOW	Allow inbound traffic from the remote computer.
*	All traffic	All	All	0.0.0.0/0	DENY	Deny all other inbound traffic.

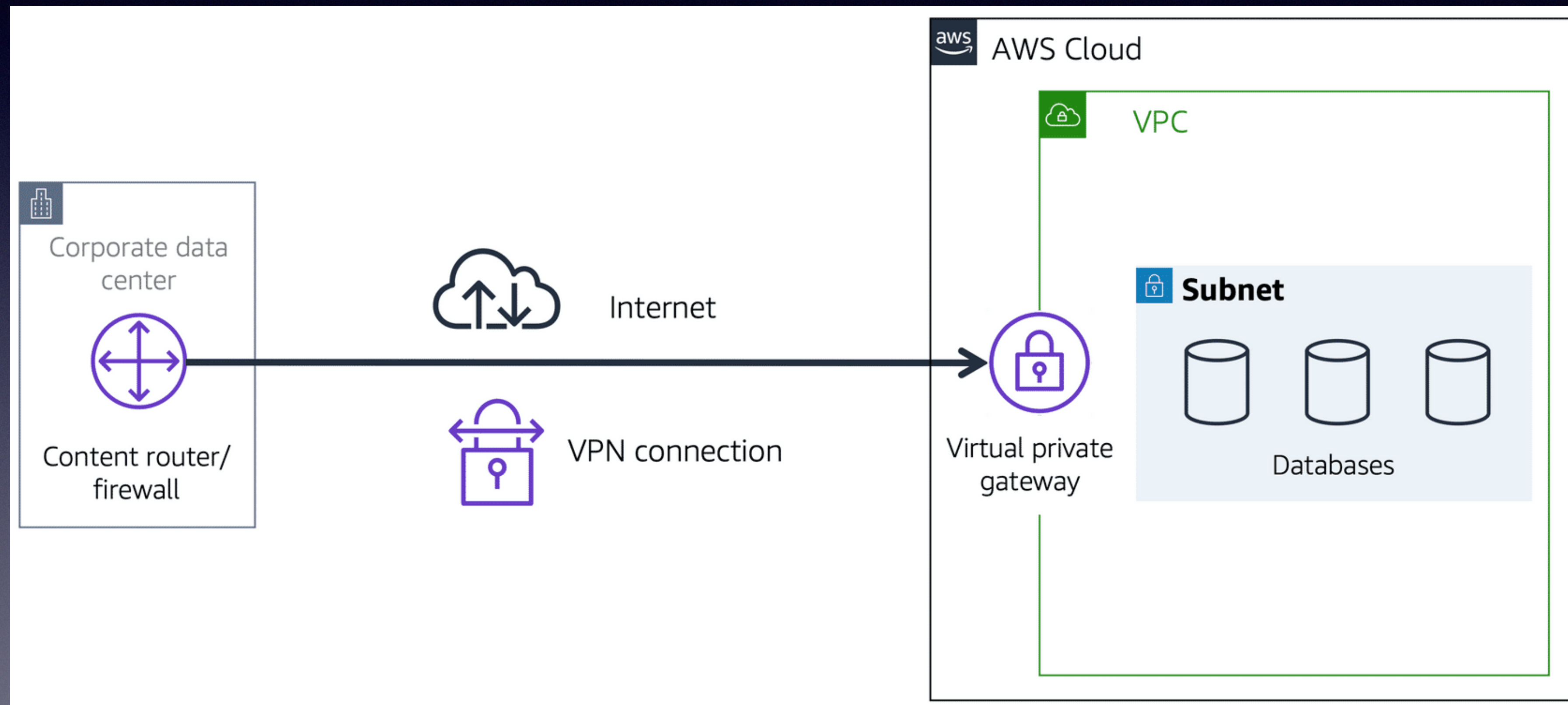
Subnets

- Subnets in a VPC control access to the Gateways and their type is determined by their routing configuration.
- A public subnet has a direct route to an internet gateway and its resources can access internet.
- A private subnet does not have a direct route to an internet gateway and its resources require a NAT device to access internet.
- A VPN-only subnet has a route to a Site-to-Site VPN connection through a virtual private gateway.
- An isolated subnet has no routes to destinations outside its VPC. Its resources can only access or be accessed by other resources in the same VPC.

Internet gateway



Virtual private gateway



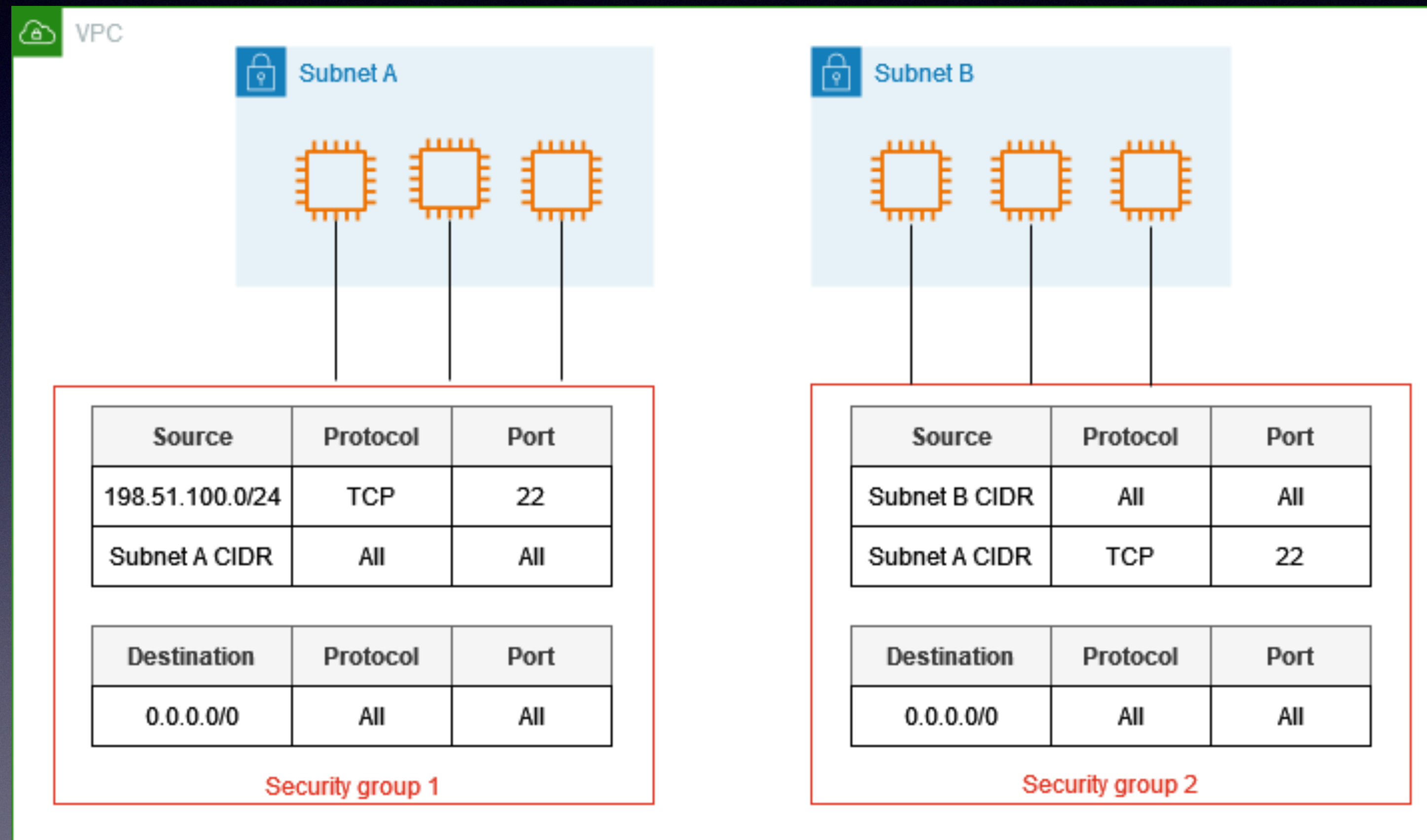
Direct connect

- Direct Connect allows you to establish a completely private, dedicated fiber connection from your datacenter to AWS.
- Low latency with high security.
- Meet high regulatory and compliance needs.
- Very expensive.

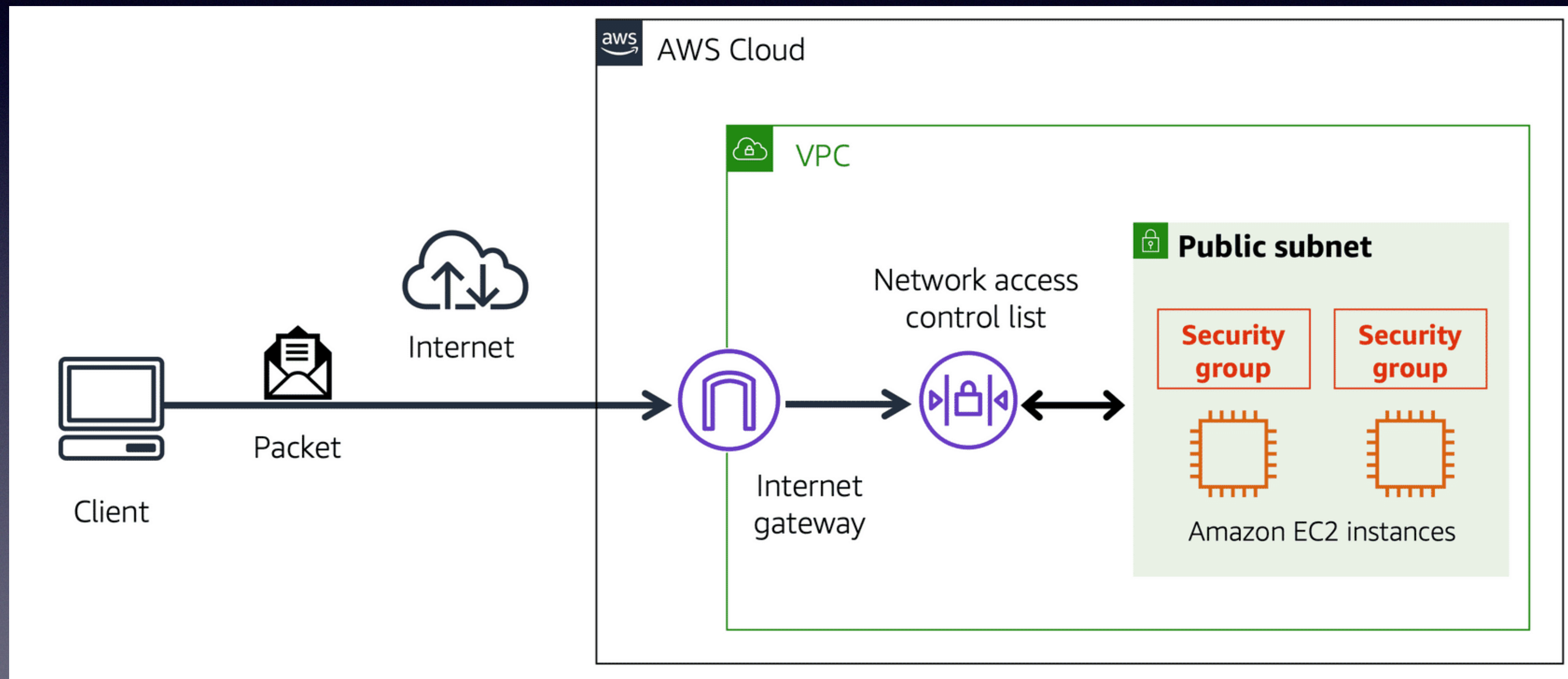
Security groups

- Every EC2 Instance is launched with a security group.
- A security group is a virtual firewall that controls inbound traffic for an EC2 instance.
- By default, the Security Group does not allow any traffic into the instance. All ports and IP addresses sending packets are blocked.
- The user can modify the Security Group to accept a specific type of traffic. For a website, accept just HTTPS requests.

Security groups example



Security checks



Storage and databases

Storing data in an EC2 instance

- An instance store volume provides temporary block storage on a disk that is physically attached to the host computer for an EC2 instance. Therefore, it has the same lifespan as the instance.
- For persistent storage, Elastic Block Store (EBS) should be used. If the instance is stopped or terminated, all the data stored on the attached EBS volume remains available.
- Before attaching the EBS volume to an instance, the user should define the configuration (such as volume size and type) of the EBS volume and then provision it. The application should be configured to write on the EBS volume.
- EBS volumes should be regularly backed up using incremental EBS snapshots.

Simple Storage Service (S3)

- S3 can store and retrieve an unlimited amount of data at any scale. Data is stored as objects in buckets.
- An object consists of data, metadata, and a key.
- Objects can also be versioned to protect them from deletion.
- Permissions limit who can see or even access objects.
- There are different storage classes that offer mechanisms for different storage use cases.

Storage classes

- S3 standard is designed for frequently accessed data. It comes with 11 nines of durability (probability to remain intact after a period of one year). The data is also stored in at least three AZs.
- S3 infrequent access provides the same level of availability as S3 Standard but with a lower storage price and a higher retrieval price. Ideal for data infrequently accessed but requires high availability such as backups.
- S3 One Zone infrequent access stores data in a single AZ and has a lower storage price than S3 Standard-IA.

Storage classes

- Intelligent-Tiering monitors objects and transfer them between standard and IA storage according to their access patterns. It requires a small monthly monitoring and automation fee per object
- S3 glacier is a low-cost storage service to archive data that have to be retained a long time, and don't need to be retrieved rapidly (few minutes to hours).
- S3 Glacier deep archive is the lowest-cost storage solution on AWS, objects can be retrieved within 12 hours.

Storage policies

- Policies can be applied on vaults in S3 glacier. For example, a write once and read many policy blocks the user from editing the data. Even the policy can be locked from future edits.
- Life cycle policies can move data automatically between tiers. For example, keep an object in S3 standard for 90 days, and then automatically move it to S3 infrequent access for the next 30 days. Then after 120 days move it to S3 Glacier.

EBS Vs S3

- S3 is web-enabled (every object has a URL) with access rights. It is regionally distributed, no need for backup strategies. It is also less expensive than EBS, with the additional advantage of being serverless.
- EBS is a block storage service while S3 is an object storage service. With Object storage, every time there's a change to the object, the entire file must be re-uploaded. With block storage only the concerned blocks are updated.

Elastic File System

- EFS is a managed file system like EBS, but it offers more features.
- EFS allows multiple instances to access its data at the same time.
- Its data can be shared between multiple applications.
- It automatically scales up and down as needed.
- Available on a region level.

Relational Database System (RDS)

- It is a managed service that automates tasks such as hardware provisioning, database setup, patching, backups, redundancy, and disaster recovery.
- RDS is available on six database engines such as MySQL, PostgreSQL, etc.
- Amazon has its own database engine called Aurora which is cost-effective and replicates six copies of the data across three AZs.

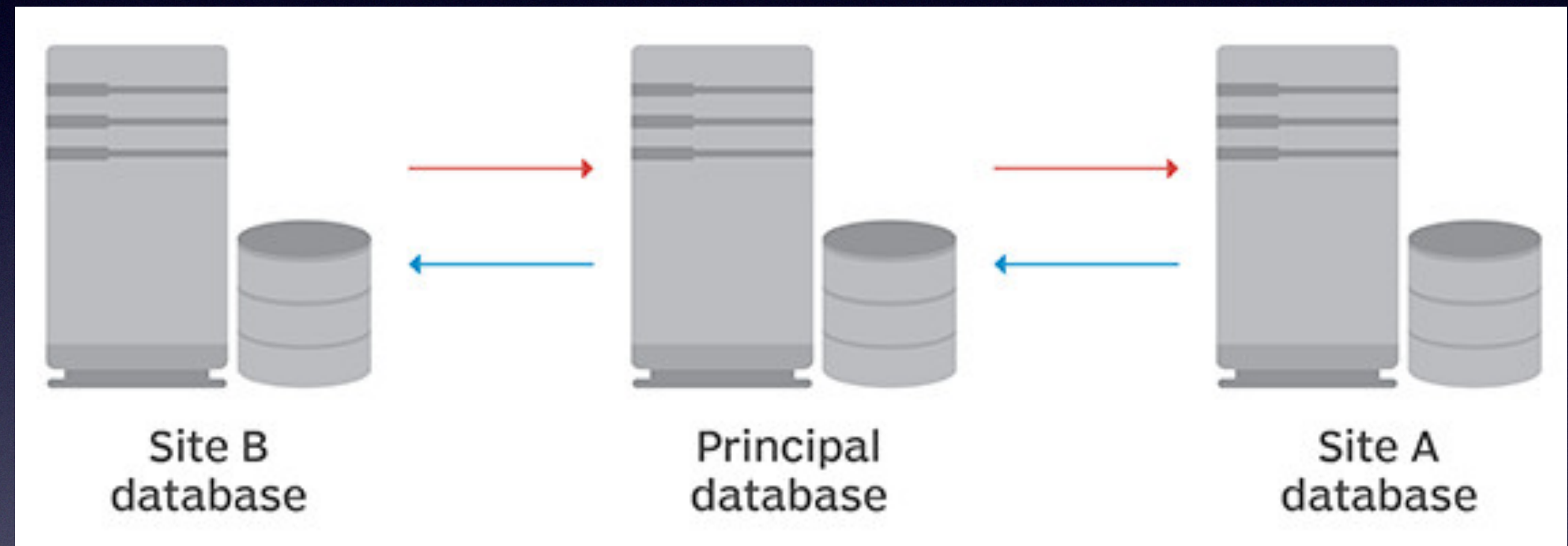
Database replication

Advantages:

- Efficiency.
- High availability.

Disadvantages:

- Data loss or inconsistency.
- Multiple servers.



DynamoDB

- DynamoDB is a non-relational key-value database (NoSQL DB).
- Data is organized into tables consisting of items and items have attributes. The tables have simple flexible schemas, attributes can be added or removed from items in the table at any time.
- It automatically scales the system up or down and stores the data redundantly across multiple availability zones.
- It is highly performant, less than a millisecond response time.

Other database services

- Other databases services: Redshift for big data, DocumentDB another NOSQL DB, Neptune a graph database, etc.
- Database Migration Service (DMS) to migrate existing databases onto AWS. It can be also used for development and test database migrations, database consolidation, and even continuous database replication.
- During the migration, the source database remains operational, reducing downtime for any applications that rely on that database.

Consolidate Databases

The main business drivers for databases consolidation are:

- **Costs** are lower with consolidation because the same number of databases run on less hardware.
- **Simplicity** is essentially another cost reduction factor because simplicity results in less labor.
- **Security** is improved by consolidation and standardization, which minimizes the number of vulnerabilities.